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Oxidation Behavior between 700 and 1300 °C of Refractory TiZrNbHfTa High-Entropy Alloys Containing Aluminum

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Refractory alloys without Cr, Al, and Si additions exhibit very poor high temperature oxidation resistance and thus significantly limit their applications. With an aim to improve the poor oxidation resistance of strong and ductile refractory TiZrNbHfTa high-entropy alloys (HEAs), this study investigates the effect of Al additions on the oxidation behavior and mechanisms for $Al_{0\text{-}1}$TiZrNbHfTa HEAs. Higher Al content renders the alloy more resistant to oxidation. The AlTiZrNbHfTa alloy exhibits a mass gain twice of that of conventional Ni-based alloys at 1100 °C for 1 h, but much less than Nb refractory alloys because an Al-containing oxide on the surface layer provides a partial barrier against oxidation between 700 and 1100 °C. But, at 1300 °C the Al-containing alloys exhibit poor oxidation resistance because the less dense oxide layers provide oxygen with an effective diffusional channel and enable oxygen to penetrate the substrate more easily.

1. Introduction

Several refractory metals and their alloys, such as Nb-based and Ta-based alloys, have the potential to be the next generation of high temperature structural materials because they have high melting points, strength at high temperatures, and ambient temperature ductility.[1–5] However, their poor oxidation resistance limits their application at elevated temperatures. Additions of Cr, Al, and Si have been shown to cause the formation of protective oxide layers of Cr_2O_3, Al_2O_3, or SiO_2.[6–8] Unfortunately at the same time, these alloying additions lead to the formation of large quantities of brittle intermetallic compounds, which significantly decrease the fracture toughness at room temperature and restricts their application.[9] Recently, Senkov et al.[10,11] proposed a new class of alloys named refractory high entropy alloys (refractory HEAs), which may offer a pathway toward addressing this problem[12–17] by employing the concepts proposed by Yeh et al. for high entropy alloys (HEAs)[18] and Cantor et al. for equiatomic multicomponent alloys.[19] HEAs are generally considered to have roughly equiatomic compositions with five or more alloy elements and to exhibit one or more solid solution phases. In addition, it has been hypothesized that severe lattice distortions caused by the variation in atomic sizes of the alloy elements also promote HEAs' mechanical properties[12,20] at both room and elevated temperatures. For these reasons, developing refractory HEAs is expected to enable researchers to find new applications in extreme environments, and this work aims to improve poor high-temperature oxidation behavior in refractory alloys.

Many studies have been carried out on refractory HEAs. In many cases, remarkably high strength at room temperature was obtained.[10,12,21–23] However, the ultimate tensile strength or fracture strength tends to be reduced at elevated temperature due to the poor oxidation resistance[5] in high temperature environments. Refractory-based alloys without Cr, Al, and Si additions exhibit very poor high temperature oxidation resistance and exhibit catastrophic oxidation[24] due to the formation of volatile oxides or non-adhering oxides. In general, avoiding or reducing the formation of high growth-stress oxides via additions of Cr, Al, and Si to the refractory alloy system ensures greater protection.[25] Al additions aim to form an alumina scale that maintains its highly protective oxide scale at temperatures above 1000 °C.[26] In addition, Al acts as a strong BCC stabilizer[12,27,28] and tends to form strong bonds with the component elements within specific refractory HEA systems, for example, TiZrNbHfTa. Thus, Al additions to TiZrNbHfTa are expected improve oxidation resistance, high temperature strength, and maintain a BCC solid solution structure.[12]

Current state-of-the-art refractory-based alloys typically rely on intermetallic phases to promote the formation of specific, protective glasses or oxides on the surface, because the underlying BCC phase exhibits poor oxidation resistance. Of the intermetallic compounds in the Mo–Si–B system, Mo_5Si_3 exhibits better creep resistance, but worse high temperature oxidation resistance than $MoSi_2$. Both porous scale formation and oxidative loss of molybdenum make Mo_5Si_3 an unacceptable candidate for high-temperature use below 1650 °C. Meyer et al.[29] showed that boron additions to the Mo–Si system promote the formation of viscous borosilicate glass, which prevents oxygen from penetrating to the substrate. This is very different from our strategies for improving oxidation resistance in refractory-based alloys. In this study, Al additions

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are employed in the TiZrNbHfTa system with an aim to improve the oxidation resistance of the inherent solid-solution phase.

Until now, few studies have investigated the mechanisms of oxidation behavior for refractory HEAs. The oxidation behavior and oxidation layer microstructure of refractory Al_xTiZrNbHfTa HEAs, where x varies from 0 to 1, are investigated in this work. We conducted oxidation experiments between 700 and 1300 °C and characterized the layered oxides via SEM and energy dispersive X-ray spectrometry (EDS). Along with the data from the oxidation experiments, the characterized oxidation layer microstructures were used to determine the oxidation mechanisms of Al_xTiZrNbHfTa alloys at temperatures between 700 and 1300 °C. New alloying strategies for improving the high temperature oxidation resistance of refractory-based HEAs are finally discussed.

2. Experimental Section

Bulk Al_xTiZrNbHfTa refractory HEAs with various quantities of Al, with $x = 0$, 0.3, 0.5, 0.75, and 1.0 - hereafter simply denoted by Al_0, $Al_{0.3}$, $Al_{0.5}$, $Al_{0.75}$, and $Al_{1.0}$, respectively, were prepared in a vacuum arc melting furnace under a high-purity argon atmosphere from raw materials with purities of 99.9 wt% or higher. Ingots were melted on a water-chilled Cu hearth. Each specimen was flipped and re-melted four times to ensure chemical homogeneity. Electrical discharge machining (EDM) was employed to cut cylindrical specimens with 8 mm in diameter and 3 mm in thickness. The final specimen exhibited a shiny surface after we ground and polished the specimens down to 1 μm diamond paste.

To evaluate the oxidation behavior of the Al_xTiZrNbHfTa alloys, we performed oxidation experiments at 700, 900, 1100, and 1300 °C in a furnace exposed to the surrounding laboratory air. These temperatures were selected because preliminary results at 500 °C showed no significant oxidation, whereas specimens tested at 1300 °C transformed completely into oxide after about 10 h (see Supporting Information). For each alloy at each temperature, six separate specimens were used to perform six different isothermal oxidation experiments for 1, 5, 10, 24, 50, and 100 h. The mass gain of the specimens was recorded, and mass gain per unit area as a function of time was used to determine the oxidation resistance of the alloys. The microstructures of the oxidized samples were investigated by scanning electron microscopy (SEM, JEOL JSM-5410). Additionally, the chemical compositions of these alloys were analyzed via EDS. Full compositional and microstructural analysis was only carried out on the $Al_{0.5}$ and $Al_{1.0}$ specimens due to time limitations for this study.

To gain insight to the oxidation mechanisms, we plotted mass gain curves on a Log-Log scale and then fit them to an oxidation growth law. The slopes of each curves were fitted over different linear time intervals to obtain the empirical oxidation growth exponents, which can reflect the oxidation growth mechanism.

3. Results

3.1. General Oxidation Behavior

The microstructures of Al_xTiZrNbHfTa refractory HEAs in the as-cast states are shown in **Figure 1**. The as-cast microstructures of Al_xHfNbTaTiZr refractory HEAs dendritic, as evidenced from the backscattered electron images (BEI). The dendrite is enriched with Ta and Nb whereas the interdendrite is enriched with Al, Hf, Ti, and Zr.[12] The oxidation behavior of Al_xTiZrNbHfTa alloys was investigated by performing isothermal oxidation experiments at various temperatures. From the appearance of the oxidized samples, the Al_0 (TiZrNbHfTa) sample exhibited pesting especially from 700 to 900 °C but also, to a much lesser extent, at higher temperatures. As shown in **Figure 2**, a large amount of oxide powder was found in the specimen holder surrounding the remaining Al_0 bulk specimen after 10 h of oxidation at 700 and 900 °C. Upon wiping off the remaining oxide on the specimen surface, we observed a rough but shiny surface on the remnant bulk. Because no

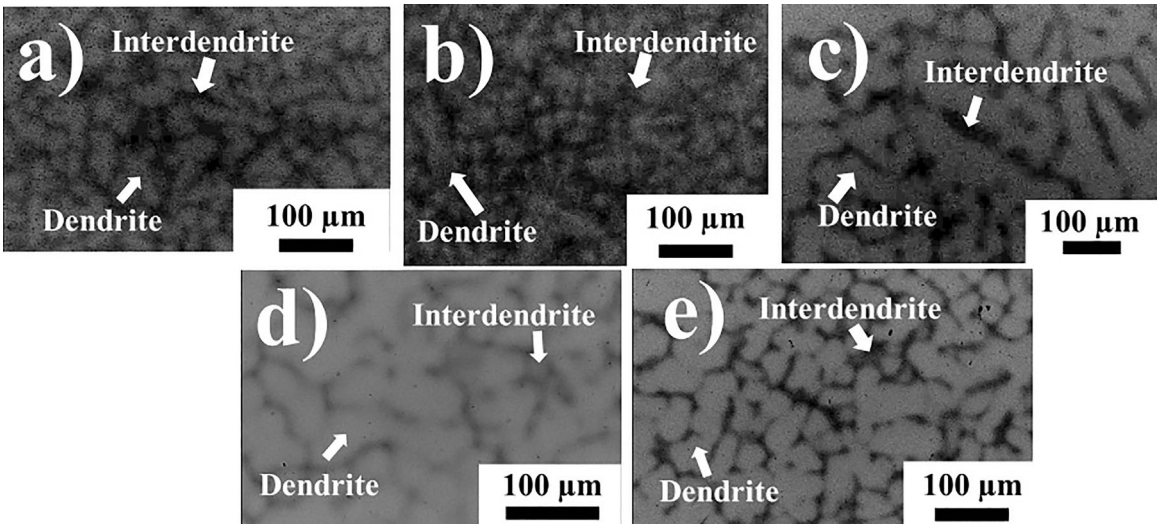


Figure 1. BEI images of Al_xTiZrNbHfTa HEAs: a) TiZrNbHfTa, b) $Al_{0.3}$TiZrNbHfTa, c) $Al_{0.5}$TiZrNbHfTa, d) $Al_{0.75}$TiZrNbHfTa, and e) AlTiZrNbHfTa alloys.

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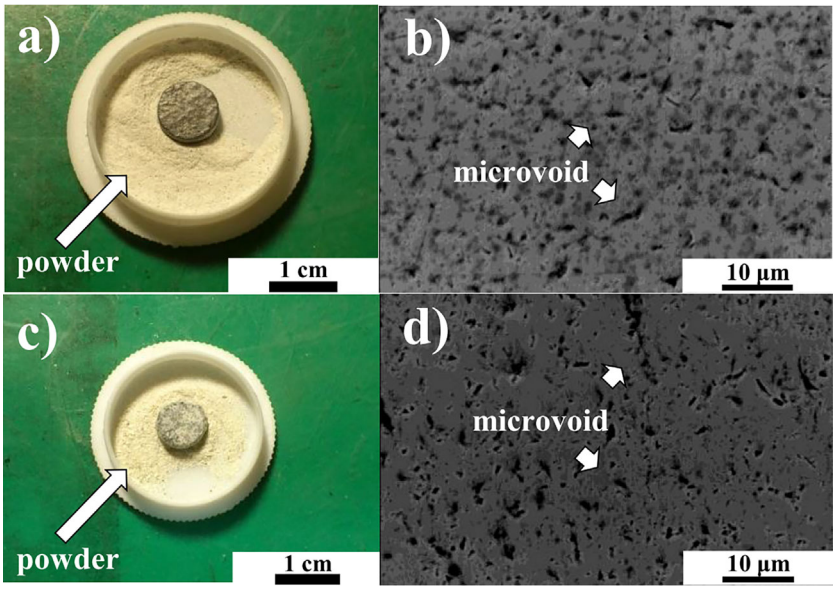


Figure 2. The pest phenomenon of the Al_0 sample (TiZrNbHfTa) at elevated temperature for 10 h: a) optical image of specimen after exposure at 700 °C and b) corresponding SEM image of the specimen surface; c) optical image of the specimen after exposure at 900 °C and d) corresponding SEM image of the specimen surface.

such phenomenon is observed in other oxidized Al-containing samples, it can be concluded that the addition of Al could be very effective at inhibiting pesting in the Al_0 alloy. The pest phenomenon involves the easy detachment of the oxidation layer formed on the surface, which causes poor protection in refractory metals and their alloys.[30] This kind of pesting in Al_0 alloy was defined as "partial pest" in contrast to the severe pesting of $MoSi_2$, which turns into powder after being oxidized for 50 h at 500 °C.[31] The explanation is that MoO_3, an oxide with high internal stress, forms below an incomplete SiO_2 layer and subsequently detaches from grain boundaries or defects in the bulk in order to release such stress.[31] Only 5 at% Al additions significantly inhibited pesting in these refractory HEAs at 700 and 900 °C, as shown in **Figure 3**. Pesting was not observed in specimens tested at 500 °C for 100 h.

Figure 4 shows the change in mass per unit area of the alloys as a function of oxidation time at 700, 900, 1100, and 1300 °C. Comparing the weight gain between various samples, we found that the specimens with a higher Al content exhibited better oxidation resistance up to 1300 °C. However, the benefits of Al additions to the oxidation resistance is particularly pronounced at lower temperatures (700 and 900 °C, Figure 4a,b) compared to higher temperatures (1100 and 1300 °C, Figure 4c,d). Further Al additions lead to the improvement of the oxidation resistance, but the improvement is gradually reduced with increasing Al content. All alloys exhibited poor oxidation resistance at 1300 °C, as shown in Figure 4d.

The mass curves in **Figure 5** were fitted to oxidation growth law with[32]:

$$\Delta m = kt^n$$

where Δm is specific mass change, k is the growth rate constant, and n is the empirical exponent used to differentiate processes limiting oxide growth. Figure 5 shows the fitting for Figure 4 on a Log-Log scale. Empirical exponents obtained from fitting are tabulated in **Table 1**. The mass gain at 700 °C in most cases follows near-parabolic growth rates except for the Al_0 specimen. The n value in the Al_0 alloy is larger than unity likely because new oxidation surfaces formed as raw materials turned into powder during exposure at 700 °C. Later, a lower n value at 700 °C after 24 h resulted from less fresh metal surface being available for pesting. All of the samples exhibited sub-parabolic/near quartic behavior at 900 °C. At 1100 °C, all of the alloys exhibited sub-parabolic/near cubic behavior, which indicates an acceleration of the oxidation process compared to that at 900 °C. At 1300 °C, a near linear behavior was observed until mass gain essentially stops from 10 h onwards. The samples that no longer exhibited mass gain (1300 °C after ≈10 h) would crumble or fall apart when handled, which indicates that the samples were completely oxidized. We were unable to perform standard metallographic analysis on these samples.

3.2. The Investigation of Oxidation Layers

Cross sections of the oxidation layer formed on the surface of the Al_0 specimen after exposure at 700 and 1100 °C are shown in **Figure 6**. The microstructure near the surface of the Al_0 alloy tested at 700 °C for 1 h is shown in Figure 6a, and it contains two

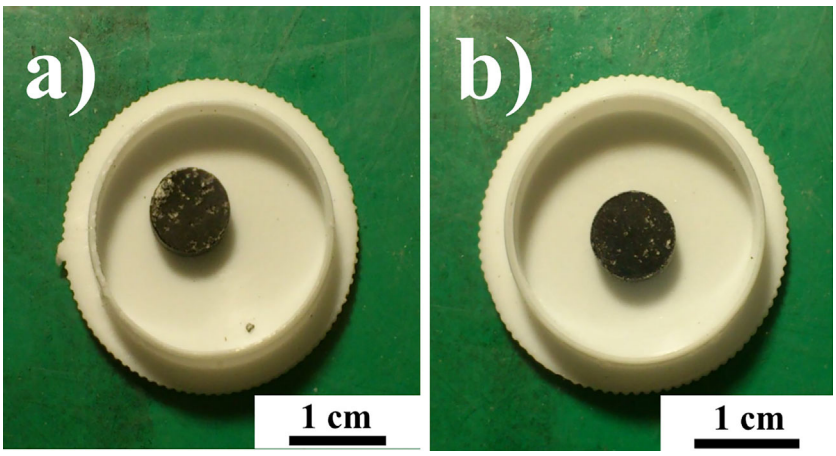


Figure 3. $Al_{0.3}$ sample ($Al_{0.3}$TiZrNbHfTa) at elevated temperature for 10 h: a) optical image of specimen after exposure at 700 °C and b) optical image of specimen after exposure at 900 °C.

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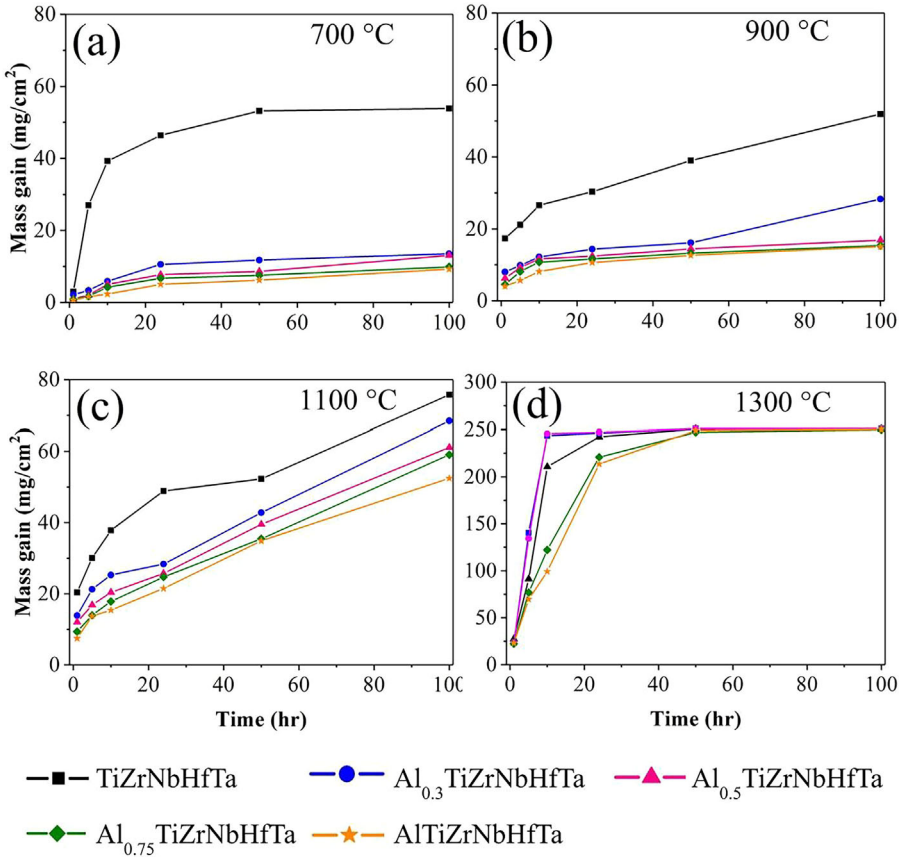


Figure 4. Comparison of weight gain due to oxidation of various Al_xTiZrNbHfTa HEAs at various temperatures: a) 700 °C, b) 900 °C, c) 1100 °C, and d) 1300 °C.

layers near the surface: the surface oxidation layer (SOL) and the oxygen diffusion zone (ODZ).[26] The oxidation layer that remains on the surface appears to be discontinuous. Figure 6b shows the surface microstructure of the Al_0 alloy exposed at 1100 °C for 1 h. The oxide layer has interfacial cracks between the SOL and ODZ, but it is still attached to the substrate in agreement with the slight pesting previously mentioned. However, based on the data shown in Figure 4a and c, we do not presume that such oxide layers can provide suitable protection at elevated temperatures. Given the higher diffusion rate of oxygen at elevated temperature and relatively low mobility of other refractory constituents, these oxide layers do not provide sufficient protection. In **Figure 7**, compositional analysis and Al-element mapping for the oxide scales of the $Al_{1.0}$ alloy indicate that both SOL and ODZ contain appreciable amount of Al and higher Al content exists in the top surface and along the interdendritic network. The enhancement of Al near the surface and along the interdendritic network relative to many of the other alloy elements may act as a partial barrier for oxygen ingress and thus improves oxidation resistance.

Figure 8 shows the cross-sectional surface microstructure of $Al_{0.5}$ and $Al_{1.0}$ HEAs tested at 1100 °C for 1 h. Both specimens exhibit interfacial cracks between oxide layers, and such stratification of oxide layers extensively appear in all samples except the ones oxidized at 700 °C. It is interesting to note that the morphology and phase composition of layers below the SOL in Al-containing samples changed with elevated temperatures (see **Figure 9**). Compared to the ODZ of $Al_{0.5}$ sample tested at 900 °C, the oxygen-rich layer (ORL) from the test at 1300 °C exhibits several phases, which include oxides and oxygen-containing BCC phases. Hence, investigating the microstructure of the ORL in these alloys is necessary to understand the effects caused by Al additions and elevated temperatures on oxidation behavior and mechanisms for $Al_{0\text{-}1}$TiZrNbHfTa HEAs.

Figure 10 shows the surface microstructure and EDS analysis of the $Al_{0.5}$ and $Al_{1.0}$ specimens exposed at 1300 °C for 1 h. The high-aspect-ratio Hf-rich phase of ORL-Hf is present within the ORL in Figure 10a. The brighter contrast of ORL-Hf indicates that it is rich in Hf and Zr and its oxide (MO_2-type) contains lower oxygen content than that (M_2O_5-type) of ORL-NbTa which is rich in Nb and Ta. ORL-Ti is dark because it is rich in Ti and contains an appreciable amount of Al, Zr, Nb, and Ta, in which Al and Ti have a smaller atomic number. It is found that the ORL-NbTa and ORL-Ti surrounds the ORL-Hf phase.

In the case of $Al_{1.0}$ HEA specimen, the stratification of oxide layers was less significant, but they could essentially be discriminated into a SOL and ORL. The ORL layer (see Figure 10b) was 150 μm in thickness with no cracks, but displayed distinctive

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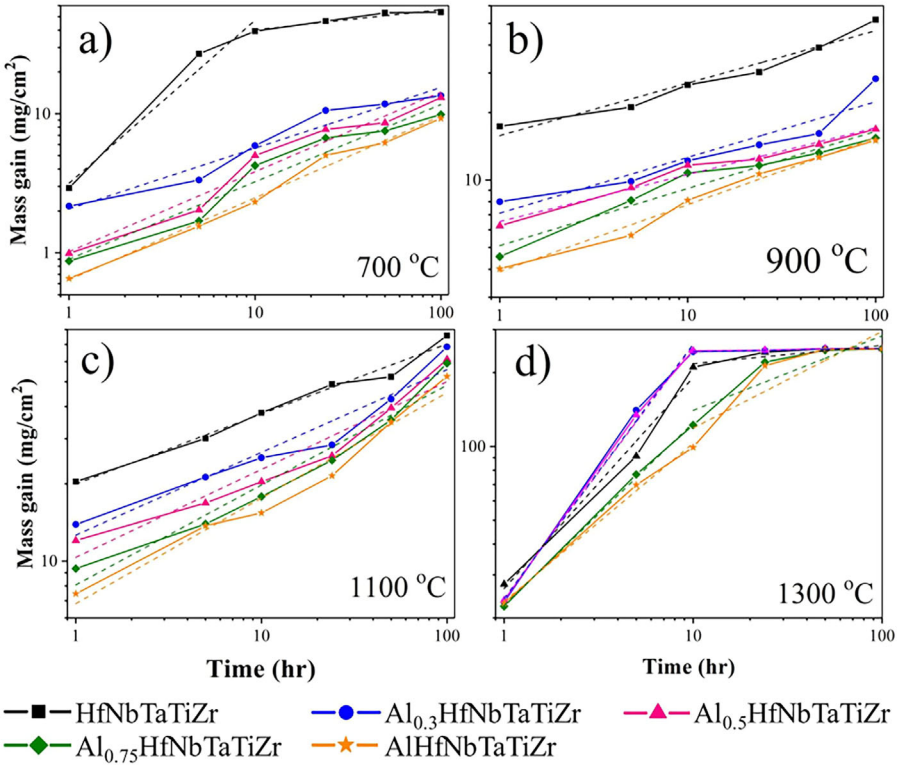


Figure 5. Comparison of Log-Log scale with fitted lines through oxidation weight gain data of various Al_xTiZrNbHfTa HEAs at various temperatures: a) 700 °C, b) 900 °C, c) 1100 °C, and d) 1300 °C.

Table 1. The fitting data of k (mg cm^{-2} s^{1})/n values for Al_xTiZrNbHfTa refractory HEAs at the exposure of various temperatures and time periods.

	700 °C		900 °C	1100 °C	1300 °C	
	1–10 h	10–100 h	1–100 h	1–100 h	1–10 h	10–100 h
Al_0	3.16/1.17	28.90/0.14	15.79/0.23	19.97/0.27	26.35/0.86	183.34/0.07
$Al_{0.3}$	2.07/0.44		7.11/0.25	12.63/0.32	23.97/1.04	236.02/0.01
$Al_{0.5}$	1.20/0.57		6.53/0.21	10.34/0.34	24.09/1.03	240.04/0.01
$Al_{0.75}$	0.90/0.56		5.09/0.26	8.07/0.39	22.53/0.74	69.70/0.30
$Al_{1.0}$	0.64/0.59		3.90/0.29	6.83/0.41	23.37/0.64	47.93/0.39

phase separation. The two dark contrast phases exhibited quite different compositions, where the ORL-Zr oxide was rich in Zr and Al with an appreciable amount of Ta, Nb, Hf, and Ti, whereas the ORL-Ti oxide was rich in Ti and contained an appreciable amount of Al, Ta, Nb, and Zr. Additionally, the bright contrast high-aspect-ratio phase, ORL-Hf, exhibited high Hf and Zr concentrations, and we believe this phase is also a MO_2-type oxide.

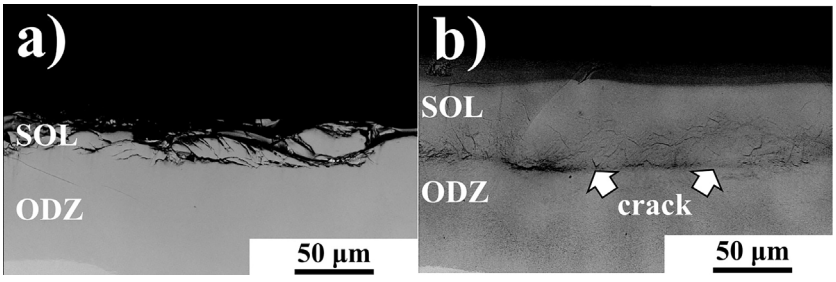
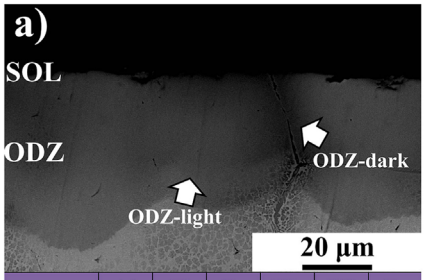


Figure 6. Surface oxidation layer of Al_0 (TiZrNbHfTa) after exposure at 700 and 1100 °C for 1 h.

4. Discussion

Figure 11 indicates that there is slight difference between the heat of formation for various oxides (–180 to −260 kcal mole^{-1} oxygen), causing several oxides to form competitively in the Al_0 alloy. Thus, we can expect that the surface oxidation layer in the Al_0 alloy is not dense to prevent oxidation, as shown in Figure 4. This fact is crucial because the alloy will not be able to avoid mass gain even when the pest

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	Al	Ti	Zr	Nb	Hf	Ta
ODZ-dark	20.9	14.2	24.6	13.0	17.5	9.9
ODZ-light	9.1	13.2	14.4	23.4	14.3	25.6

Figure 7. a) Compositional analysis and b) Al-element mapping for oxide scales of AlTiZrNbHfTa alloy treated at 900 °C for 1 h.

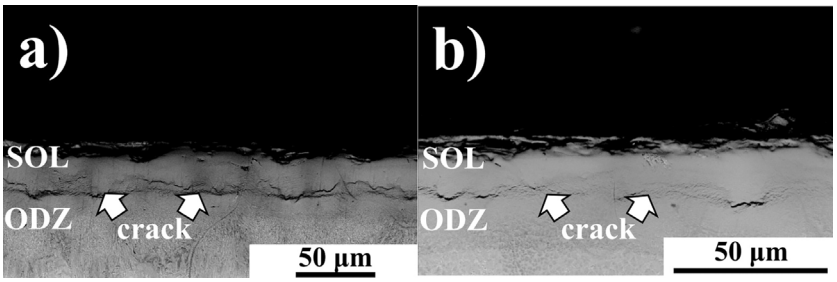


Figure 8. Surface oxidation layer of a) $Al_{0.5}TiZrNbHfTa$ and b) $Al_1TiZrNbHfTa$ at 1100 °C for 1 h.

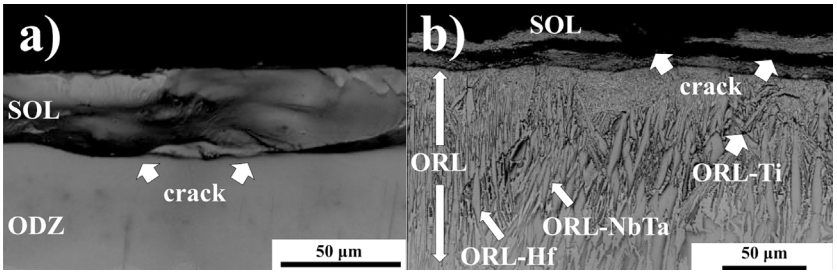
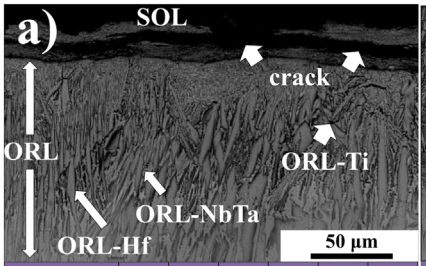
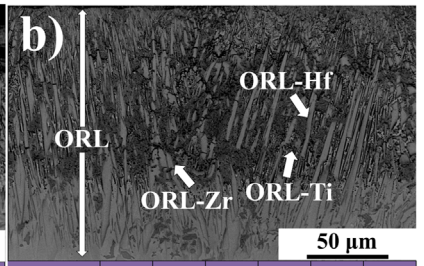


Figure 9. Surface oxidation layer of $Al_{0.5}TiZrNbHfTa$ at a) 900 and b) 1300 °C for 1 h.



	Al	Ti	Zr	Nb	Hf	Ta
ORL-Hf	1.9	5.1	23.1	7.4	59.4	3.3
ORL-NbTa	3.6	12.0	3.1	40.2	0.4	40.7
ORL-Ti	14.9	38.3	22.1	8.6	4.3	11.8



	Al	Ti	Zr	Nb	Hf	Ta
ORL-Hf	0.5	0.3	26.3	1.5	69.1	2.3
ORL-Zr	15.9	5.9	38.7	13.1	13.1	13.3
ORL-Ti	20.4	29.5	6.6	21.4	0.6	21.5

Figure 10. Surface microstructure and EDS analysis (at%): a) $Al_{0.5}TiZrNbHfTa$ and b) AlTiZrNbHfTa HEAs at 1300 °C for 1 h.

phenomenon ceases with the rising oxidation temperature. In the $Al_xTiZrNbHfTa$ system, there are two factors associated with final product in oxidation layer. First, even though the heat of formation for alumina is more negative than that for other oxides, it differs little from that for hafnia and zirconia. The distribution of the heat of formation or diffusivities of the constituent elements causes competition of oxide growth rather than only alumina forming. As a result, the oxide layer comprises of a mixture of various oxide phases. Second, according to mixing enthalpies of unlike atomic pairs (see **Table 2**), several atomic pairs possess comparatively negative ΔH_{mix}, which means that stronger bonds form between Al and other constituents (Ti, Hf, and Zr) in these alloys. Because the interdendrite region is rich in Al, Ti, Hf, and Zr and the dendrite core is rich in Ta and Nb, it is expected that oxygen species effectively diffuse along the interdendrite network and preferentially oxidize the active elements in the first stage (forming the ODZ), then penetrate into dendrite regions to oxidize the dendrite core gradually in the second stage (forming the ORL) and fully oxidize the remaining elements in the final stage (forming the SOL). Thus, three zones are formed and expanded with oxidation time (see ORL in Figure 7–9). As for the rapid oxidation at 1300 °C, it is expected that the oxygen species diffuse very fast and form oxide phases very quickly. As a result, a Hf, Zr-rich oxide grows from the interdendrite region into the dendrite to form a high-aspect-ratio-shaped phase (see Figure 10). In summary, the mixed oxides containing alumina from the initial stage provides the essential protection, and thus the higher Al content provides better protection. However, the mixed oxides are still less dense than alumina. Therefore, we could infer that little protective alumina forms during oxidation, and the formation of less dense mixed oxide layer provides moderately effective pathways for oxygen to transport and does not protect the internal metal as oxidation temperature and time increase (see Figure 4, 5).

In terms of oxidation resistance, $Al_{1.0}$ alloy shows advantages over some of pure refractory metals, Nb-based alloys and Al-free refractory HEAs.[36–42] The multiple complex oxides that are forming in Al-containing refractory HEAs provide significant oxidation resistance, as shown in **Figure 12**.[37–42] It can be seen that the mass gain of $Al_{1.0}$ alloy is much less than that of Nb refractory alloys. Furthermore, the oxidation mass gains of the present alloys are compared with those of conventional nickel-based superalloys[43–47] at various temperatures,

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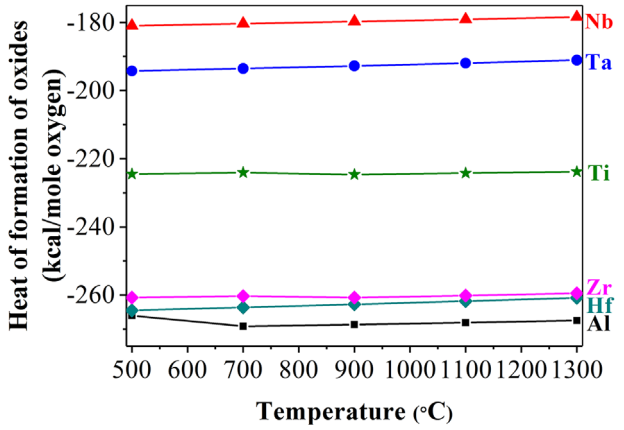


Figure 11. Heat of formation for several oxides.[33]

Table 2. Mixing enthalpies (kJ mol^{-1}) of unlike atomic pairs.[34,35]

ΔH_{mix}	Al	Hf	Nb	Ta	Ti	Zr
Al	0	−39	−18	−19	−30	−44
Hf	−39	0	4	3	0	0
Nb	−18	4	0	0	2	4
Ta	−19	3	0	0	1	3
Ti	−30	0	2	1	0	0
Zr	−44	0	4	3	0	0

as shown in **Figure 13**. The $Al_xTiZrNbHfTa$ alloy with the greatest Al addition (i.e., $x = 1$), which exhibited significant improvements in oxidation resistance, is found to exhibit twice the oxidation growth rate compared to conventional nickel-based superalloys after 1 h of exposure at 1100 °C.

Despite the poor protection at temperatures above 1100 °C, Al additions to the Al_0 HEA could lead to several advantages.[12] First, Al additions increase the yield strength and keep appreciable ductility. Second, Al additions can provide $Al_xTiZrNbHfTa$ alloys with protection from catastrophic oxidation and pesting. Even though several benefits are being presented, it is believed that suitable oxidation-resistant coatings are required to be developed for significant protection of AlTiZrNbHfTa alloys during long-term applications up to 1300 °C.

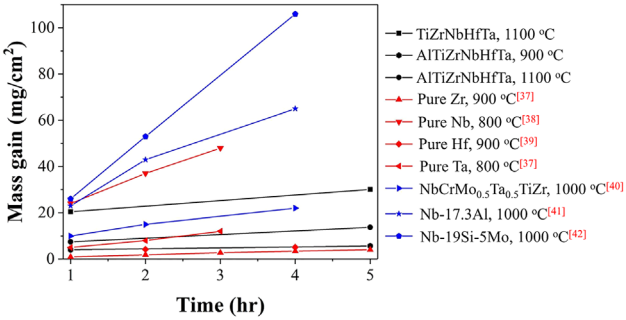


Figure 12. Comparison of pure refractory metals, Nb-based alloys and $Al_xTiZrNbHfTa$ refractory HEAs in terms of the weight gain that occurred due to 1–5 h oxidation at various temperatures.[37–42]

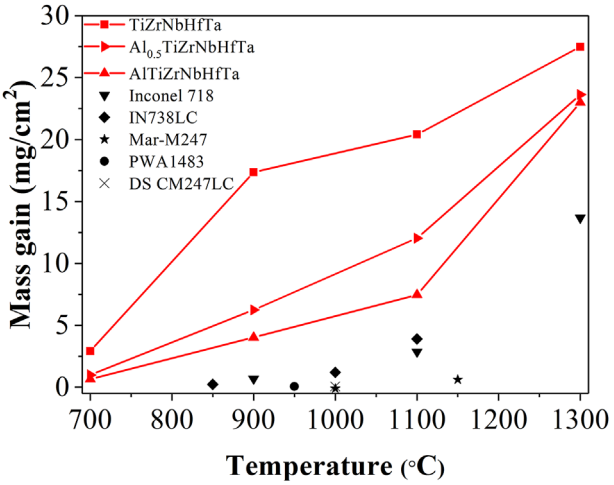


Figure 13. Comparison of conventional nickel based superalloy and $Al_xTiZrNbHfTa$ refractory HEAs in terms of the weight gain that occurred due to 1 h oxidation at various temperatures.[43–47]

5. Conclusions

The oxidation behavior and mechanisms of $Al_xTiZrNbHfTa$, refractory high-entropy alloys were studied in this work. Based on the obtained results and analysis, four conclusions can be drawn:

1) At 700 to 900 °C, pesting occurred in the Al_0 alloy and caused significant mass gain. Although such phenomenon is inhibited as the oxidation temperature increases, the mass gain of the Al_0 alloy significantly increases with increasing temperature. This indicates that the oxidation layers fail to provide the Al_0 alloy with an effective barrier against high temperature oxidation.
2) The addition of Al to the Al_0 alloy significantly improves the oxidation resistance and suppresses pesting. In the case of the $Al_{1.0}$ alloy, comparatively stable oxide scales provide an effective barrier against oxidation between 700 and 900 °C. However, the protection becomes less effective at 1100 °C and poor at 1300 °C.
3) The mass gain at 700 °C in most cases follows near-parabolic growth rates except for the Al_0 specimen. Sub-parabolic/near quartic behavior was observed in all samples at 900 °C. While at 1100 °C, all of the alloys exhibited as sub-parabolic/near cubic behavior. At 1300 °C, near linear behavior was observed until mass gain essentially stops from approximately 10 h onward in all alloys.
4) Poor protection in the Al_0 alloy is attributed to formation of mixed oxides in the oxidation layer and their low density structure. In the case of Al-containing alloys, the protection is not as good as that from dense alumina layer. It can be attributed to the formation of less dense alumina-containing mixed oxide layer because of the small difference between the heat of formation for aluminum oxide and other oxides.

Supporting Information

Supporting Information is available online from Wiley Online Library or from the author.[48]

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

Alloy design, High-entropy alloys, High-temperature oxidation, Pesting phenomenon, Refractory metals

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